

Fabian Gittins — Publications

CONTACT INFORMATION	Department of Physics and Astronomy The Johns Hopkins University, 3400 N. Charles Street Baltimore, MD 21218, United States	fgittin1@jh.edu fgittins.github.io
PUBLICATION SUMMARY	Full list of publications can be found on Google Scholar , INSPIRE-HEP and NASA ADS . h-index —16 (according to Google Scholar) or 14 (according to INSPIRE-HEP). Both include collaboration papers. Top five cited —Excluding long-author-list papers. Citation counts according to Google Scholar. 1. Gittins, F., Andersson, N., Jones, D. I., <i>Modelling neutron star mountains</i>, Mon. Not. R. Astron. Soc. 500, 5570 (2021) [arXiv:2009.12794]. (83 citations) 2. Gittins, F., Andersson, N., <i>Modelling neutron star mountains in relativity</i>, Mon. Not. R. Astron. Soc. 507, 116 (2021) [arXiv:2105.06493]. (75 citations) 3. Gittins, F., Andersson, N., Pereira, J. P., <i>Tidal deformations of neutron stars with elastic crusts</i>, Phys. Rev. D 101, 103025 (2020) [arXiv:2003.05449]. (52 citations) 4. Pereira, J. P., Bejger, M., Andersson, N., Gittins, F., <i>Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts</i>, Astrophys. J. 895, 28 (2020) [arXiv:2003.10781]. (44 citations) 5. Gittins, F., <i>Gravitational waves from neutron-star mountains</i>, Class. Quantum Gravity 41, 043001 (2024) [arXiv:2401.01670]. (37 citations)	
COLLABORATION PUBLICATIONS	Since 2026, I have been a co-author of refereed LIGO-Virgo-KAGRA and Einstein Telescope Collaboration publications. I list only non-collaboration articles below.	
SUBMITTED PUBLICATIONS	[29] Wouters, T., Ng, T. C. K., Koehn, H., Gittins, F., (3 more authors), <i>Scalable and sequential inference of the neutron star equation of state with the Einstein Telescope</i> [arXiv:2610.07975] (submitted to Phys. Rev. Lett.). [28] Yu, H., (6 authors), Gittins, F., Nanda, A., <i>Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions</i> [arXiv:2607.07943] (submitted to Phys. Rev. D). [27] Negri, L., (8 authors), Gittins, F., (6 more authors), <i>Impact of the Einstein Telescope’s duty cycle on the estimation of binary black holes parameters</i> [arXiv:2606.19201] (submitted to Phys. Rev. D).	
ACCEPTED PUBLICATIONS	[26] Gittins, F., Narola, H., Wouters, T., Pang, P. T. H., Hinderer, T., Van Den Broeck, C., <i>Detecting tidal resonances in binary neutron stars</i>, Phys. Rev. Lett. [arXiv:2606.06376].	

REFEREED
PUBLICATIONS

Research articles

- [25] Andersson, N., Counsell, A. R., **Gittins, F.**, Ghosh, S., *Tidal response of a relativistic star*, *Phys. Rev. D* **113**, 064051 (2026) [arXiv:2511.05139].
- [24] Yin, S., Andersson, N., **Gittins, F.**, *A post-Newtonian approach to neutron star oscillations*, *Class. Quantum Gravity* **42**, 235002 (2025) [arXiv:2504.06918].
- [23] Pnigouras, P., Andersson, N., **Gittins, F.**, Counsell, A. R., *Dynamical neutron star tides: the signature of a mode resonance*, *Mon. Not. R. Astron. Soc.* **542**, 1375 (2025) [arXiv:2508.06416].
- [22] Counsell, A. R., **Gittins, F.**, Andersson, N., Tews, I., *Interface modes in inspiralling neutron stars: A gravitational-wave probe of first-order phase transitions*, *Phys. Rev. Lett.* **135**, 081402 (2025) [arXiv:2504.06181].
- [21] **Gittins, F.**, Andersson, N., Yin, S., *Perturbation theory for post-Newtonian neutron stars*, *Class. Quantum Gravity* **42**, 135014 (2025) [arXiv:2503.03345].
- [20] **Gittins, F.**, Andersson, N., *Neutron-star seismology with realistic, finite-temperature nuclear matter*, *Phys. Rev. D* **111**, 083024 (2025) [arXiv:2406.05177].
- [19] **Gittins, F.**, Matur, R., Andersson, N., Hawke, I., *Problematic systematics in neutron-star merger simulations*, *Phys. Rev. D* **111**, 023049 (2025) [arXiv:2409.13468].
- [18] Counsell, A. R., **Gittins, F.**, Andersson, N., Pnigouras, P., *Neutron star g modes in the relativistic Cowling approximation*, *Mon. Not. R. Astron. Soc.* **536**, 1967 (2025) [arXiv:2409.20178].
- [17] Counsell, A. R., **Gittins, F.**, Andersson, N., *The impact of nuclear reactions on the neutron-star g-mode spectrum*, *Mon. Not. R. Astron. Soc.* **531**, 1721 (2024) [arXiv:2310.13586].
- [16] Pnigouras, P., **Gittins, F.**, Nanda, A., Andersson, N., Jones, D. I., *The dynamical tides of spinning Newtonian stars*, *Mon. Not. R. Astron. Soc.* **527**, 8409 (2024) [arXiv:2205.07577].
- [15] Beri, A., (5 authors), **Gittins, F.**, Celora, T., *AstroSat and NuSTAR observations of XTE J1739-285 during the 2019-2020 outburst*, *Mon. Not. R. Astron. Soc.* **521**, 5904 (2023) [arXiv:2303.13085].
- [14] **Gittins, F.**, Celora, T., Beri, A., Andersson, N., *Modelling neutron-star ocean dynamics*, *Universe* **9**, 226 (2023) [arXiv:2304.05413].
- [13] **Gittins, F.**, Andersson, N., *The r-modes of slowly rotating, stratified neutron stars*, *Mon. Not. R. Astron. Soc.* **521**, 3043 (2023) [arXiv:2212.04892].
- [12] Andersson, N., **Gittins, F.**, *Formulating the r-mode problem for slowly rotating neutron stars*, *Astrophys. J.* **945**, 139 (2023) [arXiv:2212.04837].
- [11] Andersson, N., **Gittins, F.**, Yin, S., Panosso Macedo, R., *Building post-Newtonian neutron stars*, *Class. Quantum Gravity* **40**, 025016 (2023) [arXiv:2209.05871].
- [10] Riley, J., (6 authors), **Gittins, F.**, (16 more authors), *Rapid stellar and binary population synthesis with COMPAS*, *Astrophys. J. Suppl. Ser.* **258**, 34 (2022) [arXiv:2109.10352].
- [9] **Gittins, F.**, Andersson, N., *Modelling neutron star mountains in relativity*, *Mon. Not. R. Astron. Soc.* **507**, 116 (2021) [arXiv:2105.06493].
- [8] **Gittins, F.**, Andersson, N., Jones, D. I., *Modelling neutron star mountains*, *Mon. Not. R. Astron. Soc.* **500**, 5570 (2021) [arXiv:2009.12794].

- [7] **Gittins, F.**, Andersson, N., Pereira, J. P., *Tidal deformations of neutron stars with elastic crusts*, **Phys. Rev. D** **101**, 103025 (2020) [[arXiv:2003.05449](#)].
- [6] Pereira, J. P., Bejger, M., Andersson, N., **Gittins, F.**, *Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts*, **Astrophys. J.** **895**, 28 (2020) [[arXiv:2003.10781](#)].
- [5] **Gittins, F.**, Andersson, N., *Population synthesis of accreting neutron stars emitting gravitational waves*, **Mon. Not. R. Astron. Soc.** **488**, 99 (2019) [[arXiv:1811.00550](#)].

Review articles

- [4] **Gittins, F.**, *Gravitational waves from neutron-star mountains*, **Class. Quantum Gravity** **41**, 043001 (2024) [[arXiv:2401.01670](#)].

Software articles

- [3] Riley, J., (6 authors), **Gittins, F.**, (16 more authors), *COMPAS: A rapid binary population synthesis suite*, **J. Open Source Softw.** **7**, 3838 (2022).

Conference proceedings

- [2] Thomas, A. Stevenson, E., **Gittins, F.**, (5 more authors), *Galactic archaeology with TESS: Prospects for testing the star formation history in the solar neighbourhood*, **EPJ Web Conf.** **160**, 05006 (2017) [[arXiv:1610.08862](#)].

WHITE PAPERS

- [1] Dietrich, T., (19 authors), **Gittins, F.**, (22 more authors), *ESO Expanding Horizon White Paper: Revealing the properties of matter at supranuclear densities with gravitational waves* [[arXiv:2512.16971](#)].